

Li Si 李斯

Full-Stack Engineer / Engineering Manager P7 · 8 YRS · JAVA-FIRST · AI-DRIVEN DELIVERY

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Eight years of full-stack engineering along two tracks: **financial-grade backend architecture** and **AI-driven development workflows**. Core developer at Webull, a 10M+ user brokerage (account compliance, in-house clearing migration, cell-based architecture). Most recently built a **15-person AI SDD team** from scratch: a Claude Code spec-driven pipeline that lifted delivery velocity **+300%** with sustained **99%** production availability, while leading a company-wide migration from Node to Java (6 → 3 months, zero major incidents). Operating principle: AI for generation, human judgment for the gates.

Core Strengths CORE STRENGTHS

SKILL · A AI SDD Development System

End-to-end Claude Code pipeline: AI-drafted templated specs → engineer review gate → implementation → automated tests with the spec as acceptance baseline; ops agent matrix; knowledge base + MCP so new hires ship in 3 days

SKILL · C High-Throughput Data Systems

Ad tracking at 6M daily / 70M peak PV: accept-only ingestion plus message-queue async processing; hot/cold attribution storage extended from 1 to 7 days; Redis cluster memory governance

SKILL · B Financial-Grade Architecture

Cell-based (unitized) systems with GZone, canary cells and a shared platform base; full account-lifecycle compliance and risk control; million-user clearing migration; 10M-scale end-of-day batch and reconciliation

SKILL · D Team & Engineering Culture

Architecture decisions, hiring and culture for a 15-person team; strangler-pattern domain-by-domain Node → Java migration with zero downtime; productivity gains measured on multiple axes

Experience EXPERIENCE

Tiger EV ENGINEERING MANAGER · TEAM OF 15

Jun 2024 – Jun 2026

EV battery-swap and battery-rental platform across Southeast Asia; unified rider / merchant / supplier product

- Owned technical direction, architecture decisions, hiring and engineering culture for a 15-person team, including the development process and quality system
- Built the Claude Code–based AI SDD development system and ops agent matrix from scratch: delivery velocity **+300%**, sustained production availability **99%**
- Led the platform-wide Node → Java migration; schedule compressed from 6 to **3 months** with zero major production incidents

Independent Consultant FULL-STACK · SOLO DELIVERY

Jan 2024 – Jun 2024

- Solely delivered the Nissan China official website (nissan.com.cn): web, Admin and API tiers, deployment and legacy data migration
- DeepSeek-based customer-service agent for a hardware-tools business (sales and order follow-up); complete backend / H5 / Admin delivery for a 400K-user health-watch product

Webull · Fumi Technology (Global R&D HQ) CORE DEVELOPER

Jun 2020 – Dec 2023

US online brokerage, 10M+ users, Nasdaq-listed; 7×24 asset-trading platform (webull.com)

- Owned account onboarding / compliance and the design of the Canada-expansion account system; responsible for statements, P&L, funding and transfers
- Integrated the in-house clearing system and migrated millions of users; designed the composite old/new-account × cell-sharding routing
- Contributed to the cell-based (unitized) re-architecture and the 7.0 gateway consolidating five client surfaces

Caohua Interactive CORE ENGINEER

Mar 2018 – Mar 2020

Unified ad-buying platform for a million-user mobile-game publisher; 30+ media integrations (Toutiao, Tencent Ads, Baidu, UC ...)

- Owned the ad-tracking and click-attribution pipeline (6M+ daily / 70M+ peak PV) and 30+ media integrations
- Built the DMP automation platform and the Hadoop / Hive marketing data warehouse from scratch

Projects PROJECTS

AI SDD Workflow & Ops Agent Matrix TIGER EV · SYSTEM DESIGNER AND DRIVER

Jun 2024 – Jun 2026

Built a spec-driven development (SDD) system on Claude Code for a 15-person team, covering requirements → spec → implementation → automated tests → operations, and turned day-to-day operations into agents.

STACK Claude Code · MCP · AI agents · Lark Open Platform · enterprise knowledge base · CI/CD

- Spec template and review gate:** specs are structured into background / technical design / API contract / test cases, drafted by AI and reviewed by engineers before any code is written — incomplete specs do not proceed; the spec doubles as the baseline for automated tests and acceptance, so "done" has an objective definition

- ▶ **Measurement framework:** lead-time comparison, iteration throughput, release frequency + rework rate, and measured case studies, cross-checked — delivery velocity **+300%**, production availability **99%**
- ▶ **Ops agent matrix** (Lark ecosystem, driven by Claude Code + MCP): a log-patrol agent that scans on schedule and pushes alerts proactively; an alert-triggered root-cause agent that reads logs, code and recent releases to produce a localization report; a support agent wired to the knowledge base for tickets
- ▶ **Knowledge base as onboarding:** reviewed specs are archived and served through MCP to both AI and people — new hires ship business features in **3 days**; rolled out to all 15 engineers

Southeast Asia Battery-Swap Platform · Full Architecture Migration **TIGER EV · MIGRATION LEAD** Jun 2024 — Jun 2026

EV battery-swap and rental platform across several Southeast Asian markets, unified rider / merchant / supplier product; migrated the early Node architecture to the Java ecosystem.

STACK Java · Spring Cloud · nginx · MQTT · message queue · MySQL · Redis · Odoo

- ▶ **Strangler pattern, not big bang:** migrated domain by domain with old and new running in parallel, business online throughout; schedule compressed from 6 to **3 months** with zero major production incidents
- ▶ **Platform architecture:** unified gateway → nginx load balancing → multi-node unified backend (swap / rental / billing / risk), horizontally scalable
- ▶ **Isolated hardware loop:** battery cabinets connect over MQTT; a hardware-sync service converges device state and task scheduler nodes run batch jobs — device-side jitter and reconnects never reach the business tier
- ▶ **Decoupled from the ERP:** async integration with the internal Odoo system over a message queue (peak shaving, replayable failures); in-house Admin centralizes multi-country, multi-tenant management and operations

Webull Account Compliance & Clearing Migration

WEBULL · CORE DEVELOPER, OWNER OF ACCOUNTS / COMPLIANCE / CLEARING Jun 2020 — Dec 2023

US brokerage (Nasdaq-listed) with 10M+ users and 7×24 trading; owned account onboarding and compliance, responsible for funds and statements, and delivered the clearing switch with a million-user migration.

STACK Java · Spring Cloud · microservices · MySQL · Redis · Kafka · end-of-day batch processing

- ▶ **Onboarding and compliance automation:** automated KYC; risk-tiered trading permissions across stocks, options and margin; a full account-lifecycle state machine with risk controls for abnormal trading and restricted lists; designed the Canada-expansion account system from scratch, handling multi-jurisdiction compliance differences
- ▶ **Statements / P&L / funding / transfers:** end-of-day batch processing for 10M+ users (daily statements and P&L); end-of-day reconciliation with clearing and banks with automatic discrepancy detection; multi-channel deposits and withdrawals (ACH, wire); automated ACATS transfers
- ▶ **Clearing migration with triple safeguards:** integrated the in-house clearing system that replaced a third party; migrated millions of users via batched canary cut-over, parallel old/new reconciliation and per-account cash and position verification
- ▶ **Composite routing design:** accounts moved with the clearing switch, so I designed the routing that layers old/new account identification on top of cell-based sharding — invisible to users, no trading interruption

Webull Cell-Based Re-Architecture & 7.0 Gateway **WEBULL · CORE DEVELOPER, CONTRIBUTOR** Jun 2020 — Dec 2023

Unitized trading architecture in the same lineage as Ant's LDC / multi-active designs, applied to a brokerage: multiple cells plus a GZone aggregation layer and a canary cell; the 7.0 gateway consolidated five client surfaces.

STACK Java microservices · Consul · unified cache · ES logging · monitoring & distributed tracing · API gateway

- ▶ **Each cell is a complete exchange:** a full microservice stack per cell (quotes / trading / accounts / risk / funding / P&L) with multi-instance services, elastic scaling, task scheduling, AZ isolation and cross-region DR; user-sharded routing keeps the blast radius inside one cell, scaling means copying a cell, and failover is a routing switch
- ▶ **Canary on real topology:** a low-spec canary cell serves a small share of traffic and designated users so releases are validated on production architecture; GZone absorbs cross-cell aggregation and global views so cells stay purely sharded
- ▶ **Shared platform base:** Consul service discovery, unified cache, unified logging / ES query, monitoring, alerting and distributed tracing — cloning a cell does not clone operational cost; contributed to the 7.0 gateway consolidating app / web / desktop / H5 / OpenAPI

Nissan China Official Website · Full-Stack Delivery

INDEPENDENT · SOLE OWNER OF API / ADMIN / WEB, DEPLOYMENT AND MIGRATION 2024

Official website of a major automaker (nissan.com.cn), delivered end to end by one engineer — from API design to production deployment, all three tiers.

STACK Next.js · React · Antd Pro · Java · SSR

- ▶ **Next.js front, Java back:** SSR secures SEO and first paint for the site (model catalog, dealer lookup, test-drive booking, news); the API service is Java, chosen for stability and engineering quality
- ▶ **Admin as CMS:** models, pages and assets are fully editable by operations, so adding content after launch never requires code — a content system, not a one-off page
- ▶ **Migration first:** legacy data exported from the old CMS, cleaned and re-imported including images and attachments; the old schema was mapped before the new one was designed

Unified ad-buying system for a million-user mobile-game publisher with 30+ media integrations (Toutiao, Tencent Ads, Baidu, UC ...), closing the loop from buying → tracking → attribution → postback → optimization.

STACK Java · MongoDB · Redis · message queue · Hadoop / Hive

- **Accept-only ingestion as the first gate:** tracking at **6M+** daily / **70M+** peak PV; endpoints answer in milliseconds and push parsing, de-duplication and attribution through a message queue to async consumers — no timeouts and no lost billing events at peak
- **Attribution retention from 1 to 7 days:** hot/cold tiering (Redis hot index + MongoDB cold store), sharded cluster with index tuning, compact document schemas, daily collections with rolling TTL — a 7× attribution window on the same resources
- **Redis keeps only the hot set:** click detail moved to MongoDB; data-structure redesign, big-key splitting and expiry policies brought cluster memory back from the alert line to a safe level
- **DMP as a closed loop, not a report:** built audience building and media sync, automated bid and budget control, conversion postbacks driving media-side oCPX and performance reporting from scratch; Hadoop / Hive marketing warehouse; buying / tracking / postback protocol differences of 30+ media unified at the platform layer

Proof of Work · Publicly Verifiable PROOF OF WORK

Webull brokerage platform 10M+ users · cell-based architecture · in-house clearing migration webull.com	Nissan China website Next.js + Java, three tiers delivered solo nissan.com.cn	Interactive resume & case files Five project dossiers (architecture diagrams / key decisions) + an LLM terminal framex-ai.com
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Tech Stack ARSENAL

PRIMARY	Java · Go · Spring Cloud · microservices · cell-based architecture · JVM & concurrency
DATA / MIDDLEWARE	MySQL · MongoDB · Redis · Kafka · Zookeeper · Consul · Quartz · Hadoop / Hive
FRONTEND / FULL-STACK	React · TypeScript · Next.js · NestJS · Hono.js · Tailwind · Antd Pro
AI ENGINEERING	Claude Code · Cursor · Codex · AI agents · MCP · SDD (spec-driven development) · DeepSeek · Docker

Education EDUCATION

Central South University (Project 985) · B.Sc. Information and Computing Science

Personal PERSONAL

Outgoing, enjoys teaching and sharing, owns outcomes rather than passing blame. Travel, photography, ultimate frisbee.